

Efficient In-Context Learning for 3D Medical Image Segmentation

Submitted to the
Albert-Ludwigs-Universität Freiburg

by Tidiane Camaret Ndir



Master's Thesis

Efficient In-Context Learning for 3D Medical Image Segmentation

Submitted on **September 28, 2026**

to the **Albert-Ludwigs-Universität Freiburg**

by **Tidiane Camaret Ndir** (5364532) [orcidlink{0009-0009-9523-2157}](https://orcid.org/0009-0009-9523-2157)

Examined by **Prof. Dr. Thomas Brox, Prof. Dr. Maxim Zaitsev**

Department of Computer Science

Faculty of Engineering

Albert-Ludwigs-Universität Freiburg

Enter your data int the Declaration.

Declaration

I hereby declare, that I am the sole author and composer of my thesis and that no other sources or learning aids, other than those listed, have been used.

Furthermore, I declare that I have acknowledged the work of others by providing detailed references of said work.

I also hereby declare that my thesis has not been prepared for another examination or assignment, either in its entirety or excerpts thereof.

Freiburg, September 28, 2026

Place, date

Tidiane Camaret Ndir

Acknowledgements

I would like to thank Robin T. Schirrmeister, Marco Reisert and Elias Kellner for their support and collaboration.

Abstract

Zusammenfassung

Contents

1	Motivation	1
2	Fundamentals	3
2.1	In-Context Segmentation	3
2.2	Volumetric Encoders and Attention	3
2.3	Synthetic Training Data	4
2.4	Related Work	4
3	Methodology	6
3.1	Problem Setup and Backbone	6
3.2	Dual Attention: Modality and Position Axes	6
3.3	Coarse-to-Fine Cascade	8
3.4	Synthetic Task Generation	8
3.5	Training Objective	9
4	Results	12
4.1	Experimental Setup	12
4.2	Image–Label Fusion	12
4.3	Coarse-to-Fine Cascade	16
4.4	Synthetic Task Generation	17
5	Discussion	19
6	Conclusion	20
6.1	Tables and References	21
6.2	Auto-complete in VSCode	21
6.3	”, Quotations and Code	22
6.4	Clipboard Manager, Git and more	23
6.5	Glossary	23
	Symbols	I
	List of Figures	II
	List of Tables	III
	List of Algorithms	IV
	Glossary	V
	References	VI

1 Motivation

In-context learning (ICL) recasts segmentation as few-shot matching: given a target scan and a small set of (image, mask) context pairs illustrating the structure of interest, a model predicts a mask for the target with no task-specific fine-tuning. This removes the need to retrain for every new anatomical class – valuable for structures that are rare, newly defined, or annotated at only a handful of sites.

Most ICL segmentation models were designed and validated in 2D. Moving to native 3D volumes is attractive – CT/MRI are inherently volumetric, and slice-wise processing discards through-plane context needed to resolve thin or elongated structures – but it sharpens an already difficult resolution/compute trade-off: dense cross-attention between a target volume and its context volumes scales with the cube of side length, so a design affordable at 64^3 becomes impractical at 256^3 .

This trade-off is not one problem but three, and this thesis addresses each with its own mechanism:

1. **How should image and label evidence be combined?** Concatenating image and mask channels before a shared encoder, or fusing them once through a single operation, commits to one fusion point before most of the network’s capacity is applied. section 3 proposes *dual attention*, keeping the two streams separate and letting them exchange information repeatedly over a modality axis and a position axis.
2. **Where should compute be spent?** Re-scanning an entire volume with a sliding window at every processing step wastes compute on regions already known not to contain the structure of interest. section 3 proposes a *coarse-to-fine cascade* that restricts fine processing to the region a coarse pass has already localized, connected across levels by a query prior and register tokens that carry a gradient path between resolutions.
3. **Where does training signal come from beyond limited annotation?** Real (image, mask) pairs are limited to the classes and appearance distribution covered by the training cohort. section 3 proposes a *synthetic task generator* that repaints real anatomical geometry with novel, controllable appearance, producing label-perfect training tasks beyond the annotated cohort.

Contributions.

- A dual attention mechanism for in-context 3D segmentation, contrasted against early-concatenation (Medverse [8]) and one-shot fusion (Iris [6]) baselines.
- A coarse-to-fine cascade in which the query prior, region-restricted fine processing, and cross-level register training are evaluated against the sliding-window-over-the-whole-volume design used by prior work.
- A GMM-based synthetic task generator with a texture-noise variant that better matches real tissue’s spatial autocorrelation, evaluated against real-only and i.i.d.-noise training.
- Evaluation on TotalSegmentator CT and a range of out-of-distribution cohorts and modalities, reporting where each mechanism helps and where it does not.

Outline. section 2 introduces in-context segmentation, the volumetric encoder and attention machinery the method builds on, and synthetic-data background, then positions UniverSeg, Iris, and Medverse against the three axes above. section 3 details the model. section 4 evaluates each axis in turn. section 5 synthesizes what the evidence supports and what remains open, and section 6 summarizes.

2 Fundamentals

2.1 In-Context Segmentation

Given a target image I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same class, in-context segmentation predicts a mask $\hat{L}^t$ for the target in a single forward pass, with no gradient update and no class-specific parameters. This sits between two more familiar paradigms: closed-set supervised segmentation, which fixes a label set at training time and must be retrained (or fine-tuned) to add a class; and interactive segmentation, where a human supplies clicks or boxes on the target image itself, per image, at inference time. In-context segmentation instead supplies the “prompt” as a handful of *other*, already-labeled examples of the same class – closer to how a radiologist might be shown a few example cases of a rare finding and asked to find it in a new scan. This is valuable in medical imaging specifically because annotation requires domain expertise, new structures and pathologies are defined faster than segmentation datasets can be built for them, and per-site protocol differences limit how far a closed-set model trained elsewhere transfers.

2.2 Volumetric Encoders and Attention

Convolutional volumetric backbones. 3D U-Net [4] and its self-configuring descendant nnU-Net [9] encode a volume through a downsampling encoder into a feature pyramid, then decode back to full resolution via a mirrored upsampling path with skip connections carrying fine detail past the bottleneck. Deeper encoder stages have coarser spatial extent and more channels; shallower stages retain spatial detail at low channel count. This split – a coarse, semantically rich bottleneck feeding a computationally light attention stack, and fine, spatially precise early stages reserved for the decoder – is exactly what the coarse/fine tap split in subsection 3.1 reuses. UNETR [7] is a prior example of combining a convolutional decoder with transformer attention for 3D segmentation.

Attention. Scaled dot-product attention [14] computes, for each query token, a weighted average of value vectors, with weights given by the similarity between the query and each key:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right) V. \quad (1)$$

Multi-head attention runs several such operations in parallel subspaces and concatenates the results, letting different heads attend to different relationships in the same input.

Position encoding. Attention itself is permutation-invariant, so position must be injected separately. Rotary position embedding (RoPE) [12] does this by rotating each query/key vector by an angle proportional to its position before the dot product, so the result depends only on the *relative* position between the two tokens rather than their absolute indices. In 3D, this is applied axially: the head dimension is split into three chunks, one per spatial axis, each rotated by an angle proportional to that axis’s physical coordinate (scaled by voxel spacing) – letting attention between two cells at different cascade resolutions still reflect their true physical distance, not just their grid index.

2.3 Synthetic Training Data

Domain randomization trains a model on procedurally varied synthetic data so it does not overfit to the narrow appearance distribution of a single acquisition protocol or training cohort. SynthSeg [1] demonstrated this at scale for medical segmentation: training pairs are generated by sampling a random Gaussian-mixture appearance conditioned only on a label map, so the resulting segmentation network never sees a real image of any specific contrast during training yet generalizes across MRI contrasts and resolutions. GIN/IPA [10] takes a complementary, appearance-only approach to single-source domain generalization: a random nonlinear intensity warp (GIN), optionally blended across spatially distinct regions of the same volume (IPA) to break spurious global appearance correlations, achieving strong generalization from randomizing appearance alone, without touching shape or geometry. Both results motivate treating appearance randomization as a distinct lever from geometric augmentation – a question the synthetic task generator (subsection 3.4) revisits directly by ablating *how* painted appearance is randomized (i.i.d. vs. spatially correlated texture noise) while holding geometry fixed.

2.4 Related Work

Three prior systems attempt in-context medical image segmentation, each making a different choice on the three axes this thesis organizes around.

UniverSeg [2] popularized the approach with a CrossBlock architecture that densely cross-attends target and context features at full resolution – no coarse-to-fine structure, and no synthetic-task training.

Iris [6] encodes each context pair into a compact per-class task embedding and fuses it into the target through a pixel shuffle-unshuffle operation – a single fusion step, applied once, with no iterative refinement across scales.

Table 1: Positioning against prior in-context segmentation models on the three axes this thesis organizes around.

	Fusion	Coarse-to-fine	Synthetic tasks
UniverSeg	dense cross-attn.	no	no
Iris	one-shot pixel shuffle	no	no
Medverse	dual U-Net cross-attn.	full-volume re-scan	no
Ours	dual attention (repeated)	region-restricted, register-carried	GMM repaint + texture

Medverse [8] encodes target and context with two separate U-Nets, fused via block-wise cross-attention between the two streams at every stage, pooling queries and keys to block resolution for efficiency (subsection 3.2).

Hierarchical, coarse-to-fine approaches. Deep Neural Patchworks [11] stacks levels of increasing resolution for large-volume segmentation, but targets single-model supervised segmentation rather than in-context, few-shot prediction – an architectural precedent for coarse-to-fine cascades outside the ICL setting. Medverse applies a similar idea within the ICL setting itself: it performs next-scale autoregressive segmentation, inspired by next-scale image generation [13], where at each step the previous step’s (perturbed) prediction re-enters as an additional context pair, processed by the same context U-Net – reused unchanged across scales. Every step still re-scans the *entire* volume with a sliding window – only the input changes with scale, not the spatial extent processed.

3 Methodology

3.1 Problem Setup and Backbone

Given a target volume I^t and a context set $\{(I_k^c, L_k^c)\}_{k=1}^K$ of image/mask pairs for the same anatomical class as the (unknown) target mask, the model predicts a target mask $\hat{L}^t$. Prediction proceeds through a coarse-to-fine cascade of M resolution levels with decreasing voxel spacing $r_1 > r_2 > \dots > r_M$; each level produces a full mask that seeds the next, finer level (subsection 3.3). Figure 1 shows our architecture (Medverse’s, for comparison, is shown in Figure 4 in the appendix).

Shared encoder. All $K + 1$ volumes of a task are passed once through a shared encoder, producing a feature pyramid per volume. Two disjoint subsets of its stages are read out: *coarse* stages, resampled to a common $R \times R \times R$ token grid ($N = R^3$ cells per volume) and consumed by the attention stack below; and *fine* stages, kept at native resolution and reserved for the decoder’s skip connections (subsection 3.2). This split keeps the attention stack on a small, fixed-size token grid while still allowing a full-resolution decoded mask.

Tokenization. Each cell of every volume yields two tokens sharing one row – an image token z^{img} and a label token z^{lbl} – assembled into a single sequence together with a small number of learned *register tokens*, shared across all volumes, that give the attention stack fixed-size read/write capacity not tied to any single spatial cell. Context rows carry the ground-truth mask in their label token; the target’s does not, since that is exactly what is being predicted – it instead starts from a neutral placeholder or, at cascade levels beyond the first, the previous level’s own prediction (subsection 3.3).

3.2 Dual Attention: Modality and Position Axes

How a model combines image and label evidence, and how context and target interact, are two related design choices this section addresses together. Medverse [8] concatenates image and label channels before a shared U-Net; context and target then interact through a Blockwise Cross-Attention Module (BAM) placed at two points in that network (once encoder-side, once decoder-side), which pools its queries and keys to block resolution before attending – trading positional precision for a cheaper attention cost – while its values stay at full resolution. Iris [6] fuses image and label features once, through a pixel shuffle–unshuffle operation, then lets context inform the target through a single bidirectional cross-attention step against a compressed, per-class task embedding computed once and reused across every query. Both commit every design choice – fusion point and context–target interaction alike – to a single, early step, computed once.

Draft diagram, ported from the conference draft; finalize labels/scale once the architecture is settled.

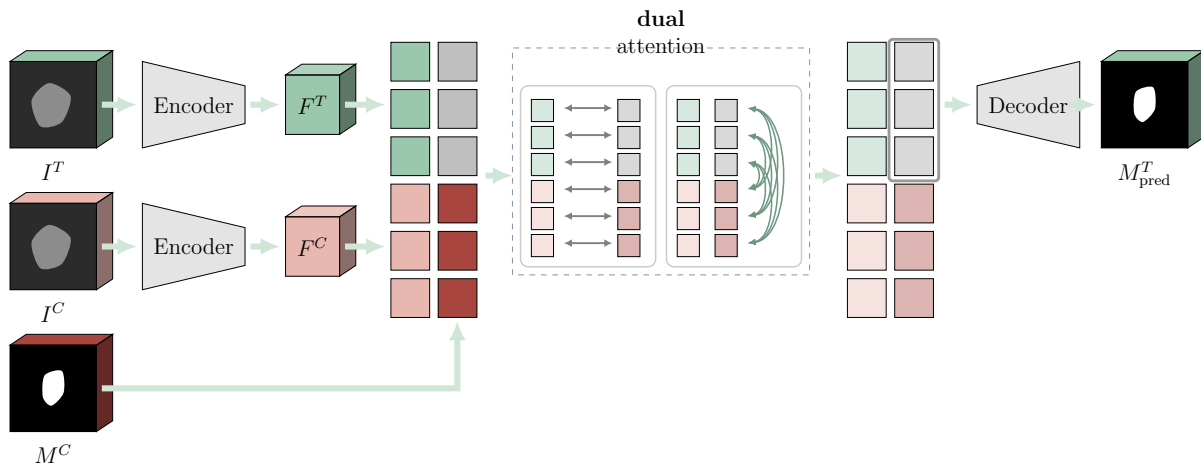


Figure 1: Architecture overview: separate image/label streams + dual attention.

We instead keep image tokens z^{img} and label tokens z^{lbl} (subsection 3.1) as separate streams throughout the network and let them exchange information repeatedly, through *dual attention* – following the general pattern of Fu et al. [5], who fuse a position-wise and a channel-wise attention branch at every layer of a scene-segmentation network – with two attention axes applied at every layer, on the full, uncompressed token grid:

- **Modality axis:** at each cell, the image and label token attend to each other, letting label evidence sharpen the image representation and vice versa, independently per cell.
- **Position axis:** every cell of every volume – context, target, and the register tokens – attends over the entire token sequence at once, using 3D axial rotary position embeddings so attention is a function of physical distance between cells rather than raw grid index. This single joint attention performs within-volume self-attention and cross-context matching together, since tokens from every volume already share one sequence.

Figure 2 illustrates both axes within a single layer.

Only the target's post-attention *image* token is read out for decoding; the label token is discarded after the last layer, since it started from a placeholder rather than real image evidence for the target row. The decoded image tokens are projected back to the coarse token grid and progressively fused with the encoder's fine-resolution skip features (subsection 3.1) via upsampling and convolution, one step per fine stage, into the final full-resolution mask logits.

Draft diagram, ported from the conference draft; finalize once the layer factorization

3.3 Coarse-to-Fine Cascade

The cascade connects consecutive resolution levels through three mechanisms, each replacing a coarser design choice made by Medverse or Iris.

Query prior. Medverse [8] feeds an upsampled, perturbed ground-truth mask into an extra U-Net branch at every step; Iris [6] uses no prior at all. At level $\ell > 1$, we instead initialize the target’s label token directly from level $\ell-1$ ’s own prediction:

$$z_{\ell,0}^{\text{lbl}} = \text{LblEmbed}(\text{Warp}_{\ell-1 \rightarrow \ell}(\hat{L}_{\ell-1}^t)), \quad (2)$$

where Warp resamples level $\ell-1$ ’s prediction onto level ℓ ’s crop grid, correcting for the two levels’ different centers, field of view, and any augmentation applied independently to each. This makes the previous level’s belief part of the same token stream processed by dual attention alongside the image (subsection 3.2), rather than a separate side channel. Figure 3 shows this flow across two levels.

Region-restricted fine processing. Medverse and Iris re-scan the entire volume with a sliding window at every step. Once a coarse level has localized the structure of interest, we instead restrict the sliding window at the next, finer level to the coarse prediction’s foreground region, thresholded and padded by a fixed margin. Compute at fine levels then scales with the size of the structure rather than the size of the input volume, at the cost of depending on the coarse level’s localization being approximately correct.

3.4 Synthetic Task Generation

A fraction p_{synth} of training tasks are drawn from a synthetic generator instead of real context/target subjects, broadening training beyond the anatomical classes and appearance distribution the real cohort covers.

Procedural shape generator. Each synthetic task samples a binary target mask from one of seven procedural shape families (blob, splatter, disk, cylinder, scatter_field, vessel, torus) and paints it onto a canvas image. Paint intensity is host-anchored – drawn relative to the canvas region’s own local mean/std rather than an absolute value – and textured with multi-octave, spatially correlated noise (matching real tissue’s autocorrelation, unlike flat i.i.d. paint), with an optional core/rim intensity blend for added heterogeneity. *scatter_field* specifically targets lesion *multiplicity* (many small, spatially scattered components), a pattern most anatomical structures in the real training cohort do not exhibit.

Draft diagram, ported from the conference draft; finalize once the number of levels M is settled.

Canvas: synthetic bank vs. real subject. Two canvas sources are available. The *synthetic-bank* canvas draws a candidate mask/image region from a precomputed bank of segmented real scans and paints the shape into it. The *real-host* canvas instead paints the same procedural shape directly onto a real training-cohort subject’s own image (CT or MRI, any organ), bypassing the bank entirely: every volume in a task shares one real subject as its base image, differing only in the painted region.

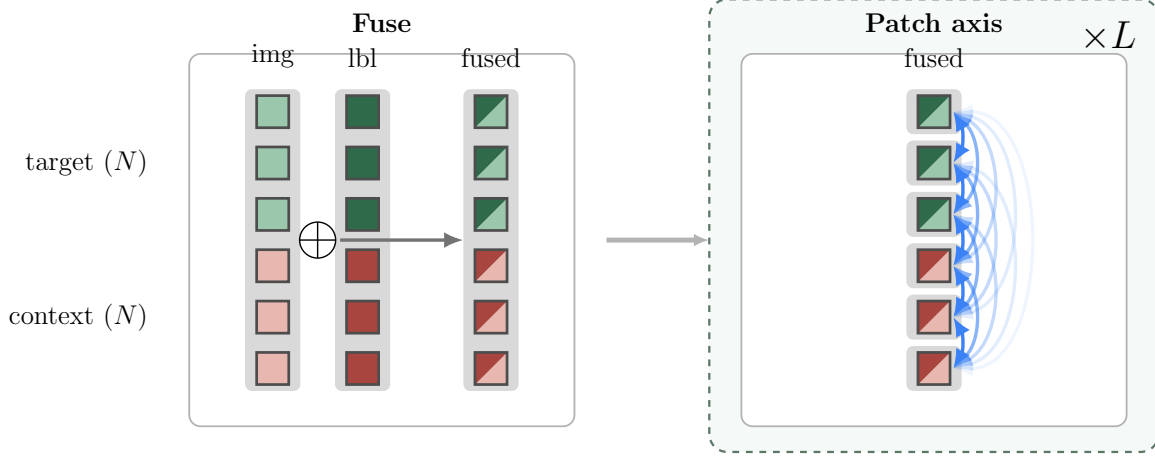
3.5 Training Objective

Each cascade level is supervised independently at its own resolution with a combined binary cross-entropy and soft Dice loss, summed over the M levels:

$$\mathcal{L} = \sum_{\ell=1}^M \lambda_{\ell} (\mathcal{L}_{\text{BCE}}^{\ell} + \mathcal{L}_{\text{Dice}}^{\ell}) . \quad (3)$$

Level weights λ_{ℓ} are uniform ($\lambda_{\ell}=1$) in the reference configuration; the query-prior path is detached between levels (subsection 3.3), so each level’s loss trains its own prediction without back-propagating through the coarser level’s mask.

Baseline: early fusion



Ours: dual attn.

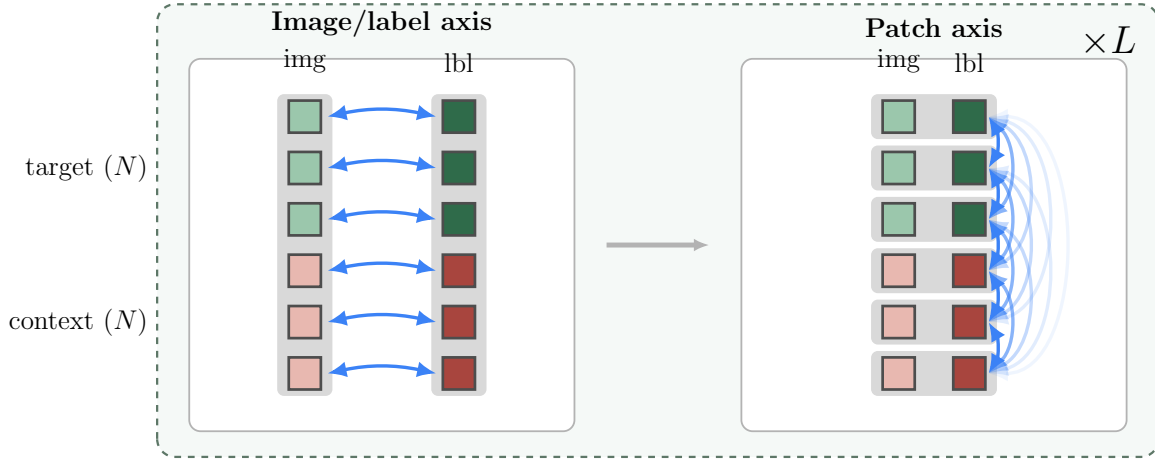


Figure 2: Dual attention within one transformer layer, shown for a single context volume ($K=1$ for clarity) and 3 of its N spatial tokens per stream. Left: the modality axis – image and label tokens attend to each other within the same cell. Right: the position axis – tokens attend across cells and, for the target, across contexts. Green edges touch the target row, which cross-attends read-only into context and whose label-stream input is the query prior (subsection 3.3) rather than ground truth.

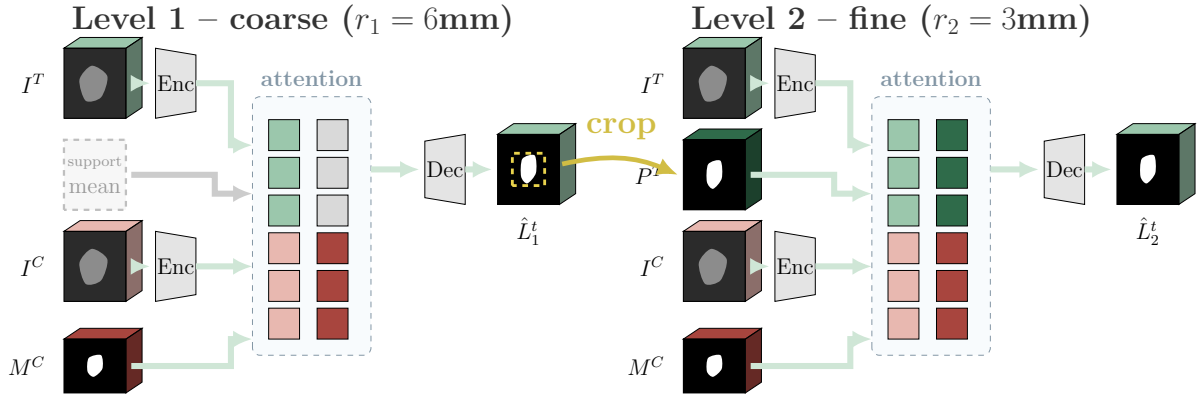


Figure 3: Query image prior across two cascade levels, with the full model input/output flow. At level 1 the target has no real prior yet (dashed support-mean placeholder); at level 2 the target's second lane is a real crop of level 1's own prediction $\hat{L}_1^t$ – the same lane the context mask always occupies.

4 Results

4.1 Experimental Setup

Model scope. Two configurations of the architecture are reported in this chapter, kept explicitly separate. The reference configuration (dual attention, query-prior init, register carry gated off) is the source of every Fusion and Cascade number below. An extended configuration (adding cascade registers, a pooling token, and synthetic texture noise on top of it) is the source of every Synthetic Task Generation number. The two are trained independently and not directly comparable to each other – the extended configuration is read as a promising newer result, not a replacement for the numbers reported above it.

Baselines. Medverse [8] (released weights, and – where noted – finetuned on the same TotalSegmentator split) is the only baseline evaluated empirically here.

Metrics. Accuracy is reported as Dice (macro, per-class then averaged, unless stated per-sample) and Normalized Surface Dice (NSD). Compute is reported as GFLOPs, wall-clock latency, and peak VRAM where available; VRAM comes from a standalone $B=1, K=1$ inference benchmark rather than the evaluation loop itself, so it should not be read as representative of a real $K>1$ eval run.

Datasets. Table 2 lists every evaluation source used in this thesis, grouped by relationship to the training vocabulary. TotalSegmentator (the in-distribution source) and FLARE22 both reuse the training vocabulary [15], splitting into a genuine seen/unseen pair; the rest introduce a target, class, or scanner modality absent from training entirely.

*Sourced from an unofficial data mirror; provenance not yet independently confirmed.

4.2 Image–Label Fusion

Two comparisons follow, each isolating one design choice. The first asks whether moving from Medverse’s dual U-Net and cross-attention to a shared encoder with attention already helps, before anything else about this thesis’s architecture is considered; the second asks whether keeping image and label evidence as separate streams – dual attention – helps further, within that same encoder-and-attention family.

Setup. Table 1 classifies Medverse’s fusion strategy as dual U-Net cross-attention: target and context are encoded by two separate U-Nets and fused via cross-attention between the two streams at every stage, rather than through a single shared encoder. This arm is labeled **Dual U-Net Cross-Attn** throughout, not Medverse: it starts from

Resolve before submission: does the thesis report a single “Ours” configuration, or keep this split?

No Iris baseline exists yet; the comparison in subsection 2.4 remains architectural, not empirical.

Add citations for each source

Table 2: Every evaluation source, grouped by relationship to the training vocabulary. n is subjects; n_{cls} is evaluated classes.

Source	Modality	Target	n	n_{cls}
<i>In-vocabulary (seen/unseen split)</i>				
TotalSegmentator	CT	117 organs (61 seen / 56 unseen)	89	117
FLARE22	CT	abdominal organs (11 seen / 2 unseen)	50	13
<i>Novel vocabulary (unseen by construction)</i>				
ISLES22	MRI DWI	stroke lesion	250	1
Shifts-MS	MRI FLAIR	MS lesion	46	1
MSD Hippocampus	MRI T1	hippocampus	260	2
MSD Prostate	MRI T2+ADC	prostate zone	64	2
ATLAS v2.0*	MRI T1	chronic stroke lesion	654	1
GNC_705	MRI Dixon	kidney lesion	610	13
NasalSeg	CT	nasal/sinus cavities	107	5
<i>Non-object</i>				
HU_LWK1	CT	L1 vertebra ROI	36	1
<i>Held-out modality</i>				
TotalSeg MRI	MRI	same classes as TotalSegmentator CT	47	23

Medverse’s released weights (the best available starting point for this architecture) but is then trained under the identical recipe as the other two arms for the same ~ 250 epochs – a different thing from the released, unmodified baseline evaluated elsewhere in this thesis (subsection 4.3). The single-axis control isolates the architectural family this comparison targets: it keeps the shared encoder and attention stack common to every architecture in this thesis, but collapses image and label evidence into one token per cell via a pixel shuffle–unshuffle operation *before* the transformer – echoing Iris’s own fusion mechanism in spirit though not in implementation, and withholding dual attention until the next step of the staircase.

Its transformer depth is raised from $l=4$ to $l=9$ specifically to compute-match the dual-attention arm (removing the modality axis also removes a large share of the model’s compute; parameter count and compute could not be matched simultaneously, leaving parameters 52% higher for the single-axis control). This adjustment is made relative to dual attention only, not to Dual U-Net Cross-Attn, so the Dual U-Net Cross-Attn comparison alone is not compute-matched.

All three arms train under an identical recipe (Table 3), an identical data pipeline, and an identical augmentation recipe, but not an identical starting point: Dual U-Net Cross-Attn keeps its released initialization, while the single-axis and dual-attention arms warm-start only the encoder and decoder from a shared checkpoint, with the

Table 3: Training settings shared by Dual U-Net Cross-Attn, the single-axis control, and dual attention (ours). Epoch count is a placeholder pending confirmation of the exact per-arm value.

Optimizer	AdamW
Learning rate	10^{-4}
Schedule	cosine decay, 10-epoch warmup
Weight decay	10^{-2}
Batch size	8
Gradient clipping	1.0
Loss	BCE + Dice
Epochs	~ 250

entire in-context reasoning core (transformer, image/label embeddings, context/query id, register tokens, cascade projection, pool projection) randomly initialized for both – so neither patchset arm starts from an already-trained transformer. Training uses real anatomy only ($p_{\text{synth}}=0$), CT only. The three arms also did not train to a matched stopping point: Dual U-Net Cross-Attn trained to full convergence, the patchset arms did not.

All three arms are then evaluated with an identical protocol – fixed 3 mm single-level crops, fp32, each model’s own best checkpoint – across every dataset this thesis evaluates (subsection 4.1), not TotalSegmentator alone.

Rows below are grouped by each dataset’s relationship to the training vocabulary, not by how different the source otherwise looks: TotalSegmentator itself (61 seen / 56 unseen classes) and FLARE22 (11 seen / 2 unseen, out of 13 total, subsection 4.1) both reuse the training vocabulary and so split into a genuine seen/unseen pair; MSD Prostate and MSD Hippocampus introduce organs with no vocabulary overlap at all, so every class in that pair is unseen by construction, not by a train/val split; HU_LWK1’s target is a HU-measurement region, not a segmentable anatomical object, a categorically different kind of “unseen” from a novel organ; TotalSegmentator’s own MRI acquisition isolates a held-out *modality* instead, over classes the vocabulary already contains.

Results. Single-axis alone already beats Dual U-Net Cross-Attn on five of the eight rows – both TotalSegmentator splits, both FLARE22 splits, and TotalSeg MRI – while Dual U-Net Cross-Attn wins the three remaining rows: MSD Prostate, MSD Hippocampus, and HU_LWK1. Restoring dual attention flips one of those three (MSD Hippocampus, 0.184 vs. 0.172) without touching the other two, bringing the full architecture to six wins out of eight against Dual U-Net Cross-Attn. The one aggregate number in the table – mean of the five held-out sources – shows why that flip matters: single-axis alone is a net regression against Dual U-Net Cross-Attn there (0.264 vs. 0.278), and only once

Table 4: Dual U-Net Cross-Attn, the single-axis control, and dual attention (ours), across all evaluation datasets (subsection 4.1). Grouped by vocabulary relationship: in-vocabulary seen/unseen, novel-vocabulary (unseen by construction), non-object, held-out modality.

	Dual U-Net Cross-Attn	Encoder + Attention	
		Single-axis	Dual attention (ours)
TotalSeg CT – seen	0.348	0.574	0.582
TotalSeg CT – unseen	0.221	0.408	0.426
FLARE22 – seen	0.532	0.693	0.709
FLARE22 – unseen	0.443	0.632	0.604
MSD Prostate (unseen)	0.377	0.205	0.247
MSD Hippocampus (unseen)	0.172	0.122	0.184
HU_LWK1 (non-object)	0.146	0.079	0.102
TotalSeg MRI (modality)	0.178	0.232	0.259
mean of 5 held-out sources	0.278	0.264	0.297
l	–	9	4
params	71.1M	74.1M	48.7M
GFLOPs	2362.6	3973.1	3892.5
latency (fp32, per sample)	73.2 ms	73.0 ms	72.5 ms

dual attention restores the modality axis does the full architecture overtake Dual U-Net Cross-Attn (0.297). The architectural family change alone, in other words, is not what carries this result; dual attention is. The compute picture is mixed rather than uniformly favorable: dual attention has the fewest parameters and the lowest latency of the three, but not the fewest GFLOPs – Dual U-Net Cross-Attn remains cheapest by that measure despite its lower accuracy, so no arm here is a strict win on both accuracy and compute at once. Two caveats keep this from a single-variable ablation: Dual U-Net Cross-Attn starts from a pretrained initialization rather than the randomly-initialized reasoning core the patchset arms share, and does not receive the same query-conditioning signal during training – so this is best read as a comparison against Dual U-Net Cross-Attn’s strongest available starting point, not a fully symmetric ablation.

Status. Neither patchset arm had reached its full training budget at the point these numbers were read, while Dual U-Net Cross-Attn had already converged – so Table 4 may understate Dual U-Net Cross-Attn’s eventual ceiling more than either patchset arm’s. All three arms are N=1 seed, and several rows rest on very few classes (FLARE22’s unseen split, both MSD sources, HU_LWK1) – directional evidence, not settled effect sizes.

Re-read once
Dual U-Net Cross-Attn receives the same query-conditioning signal – currently dropped

Table 5: Predicted-prior checkpoint vs. Medverse (native autoregressive, depth-matched), macro Dice and mean per-sample latency. Levels/spacing is our own cascade ladder for that dataset; Medverse’s AR depth is matched to the same level count.

Dataset	Levels	Spacing (mm)	Dice		Latency	
			Ours	Medverse	Ours	Medverse
TotalSeg CT	3	[6, 3, 1.5]	0.586	0.103	278ms	4978ms
FLARE22	3	[6, 3, 2.5]	0.751	0.359	311ms	4894ms
MSD Prostate	3	[3, 1.5, 0.75]	0.375	0.357	279ms	5006ms
MSD Hippocampus	1	[0.5]	0.303	0.686	99ms	50ms
HU_LWK1	3	[6, 3, 1]	0.152	0.020	439ms	5336ms
TotalSeg MRI	3	[6, 3, 1.5]	0.506	0.038	306ms	4842ms
NasalSeg	2	[1.5, 0.8]	0.547	0.737	161ms	1886ms
mean			–	–	268ms	3856ms

4.3 Coarse-to-Fine Cascade

This section asks whether the coarse-to-fine cascade introduced in subsection 3.3 helps in-distribution at all; how the resulting design compares to Medverse’s own coarse-to-fine mechanism, on accuracy and compute; and which of the cascade’s own mechanisms are responsible for the in-distribution gain.

Coarse-to-fine cascading improves accuracy in-distribution. A spacing sweep from 4mm to 1.5mm crop resolution (117 classes, 3897 samples) raises macro Dice from **0.361 to 0.538** (+0.177). The locator’s containment of the target structure is already high at the coarse level (0.922 mean, vs. an oracle ceiling of 0.970 – a 0.048 gap) and does not predict which classes gain the most at the fine level: once a class is roughly localized, the remaining error is dominated by segmentation quality at the fine crop, not by further localization failure.

Ours vs. Medverse, seven datasets, accuracy and compute. “Ours” here is the predicted-prior checkpoint (the best-performing configuration of the query-prior ablation below); “Medverse” is released weights, native autoregressive inference – its own coarse-to-fine mechanism – with its refinement depth level-matched to our own cascade ladder per source – the fairer comparison, even though matching that depth actually *hurts* Medverse’s own accuracy. Four held-out sources (ISLES22, Shifts-MS, ATLAS v2.0, GNC_705) are excluded – both models score too low there to be informative. Table 5 reports macro Dice and mean per-sample latency for both.

Ours wins 5/7 (TotalSeg CT, FLARE22, MSD Prostate, HU_LWK1, TotalSeg MRI); Medverse wins 2/7 (MSD Hippocampus, NasalSeg) – both cases where the target is

TotalSeg
CT’s
Med-
verse
row
is not
yet the

Table 6: Query-prior source ablation, macro Dice, across all validation datasets. TotalSeg CT also reports NSD (0.576 / **0.631**).

Dataset	GT-prior (perturbed)	Predicted prior (real prev. pred)
TotalSeg CT	0.543	0.586
FLARE22	0.731	0.751
MSD Prostate	0.310	0.375
MSD Hippocampus	0.319	0.303
HU_LWK1	0.099	0.152
TotalSeg MRI	0.449	0.506
NasalSeg	0.399	0.547

a single, large, well-defined structure that suits Medverse’s full-FOV multi-resolution pyramid. Compute favors Ours by roughly $14\times$ on mean latency – and this is *after* level-matching Medverse’s own depth to ours, which costs Medverse accuracy too, so an unmatched Medverse would lose by even more on both axes at once. Medverse’s own coarse-to-fine mechanism is separately prior-sensitive: feeding its own imperfect coarse prediction forward as a query prior hurts its harness cascade universally, across all 7 OOD sources tested, sometimes catastrophically (MSD Hippocampus $0.691 \rightarrow 0.050$) – compounding coarse-level error appears to be a property generic to coarse-to-fine cascades, not specific to either architecture’s design.

A real predicted prior beats a perturbed ground-truth prior. Same checkpoint chain, same seven validation datasets as Table 5 plus in-distribution TotalSeg CT (Table 6):

Feeding the model’s own real previous prediction forward beats a perturbed ground-truth prior on 99 of 117 classes (median $\Delta + 0.043$) – a *prior-source* ablation (real vs. noisy synthetic), distinct from the Medverse *prior-presence* finding above; a true no-prior arm is still needed (gap G6). Out-of-distribution (9 held-out sources including FLARE22/NasalSeg): the prior-source gain helps 5/9 sources and hurts 4/9 – weaker than the clear in-distribution majority.

4.4 Synthetic Task Generation

On four held-out sources with no anatomical overlap with training (ISLES22, Shifts-MS, ATLAS v2.0, GNC_705 – lesion targets on MRI, one Dixon), the baseline checkpoint scores Dice 0.008–0.082. Widening the shape generator’s family mix or dosage on the synthetic-bank canvas (subsection 3.4) left every source within noise of baseline. The bank turned out to be built entirely from CT scans, while every target source here is MRI – a canvas-modality mismatch, not a shape-diversity problem.

Add a no-query-prior control arm to complete the ablation table promised in the method.

Table 7: OOD Dice, real-host canvas progression. Stage 1→2 is a single-variable comparison (canvas only); stage 2→3 additionally extends to all shape families, a wider real-subject pool, and a longer budget.

Stage	ISLES22	Shifts-MS	ATLAS v2.0	GNC_705
Baseline	0.050	0.008	0.028	0.082
1) Synthetic-bank canvas	0.045	0.011	0.033	0.070
2) Real canvas, one family	0.098	0.058	0.032	0.105
3) Real canvas, all families	0.087	0.051	0.027	0.223

Table 8: TotalSegmentator validation Dice for the same checkpoints. seen/unseen: macro Dice by trained-class membership. MRI/CT: Dice by target modality.

Stage	Dice	seen	unseen	MRI	CT
Baseline	0.418	0.489	0.344	0.390	0.468
1) Synthetic-bank canvas	0.422	0.485	0.356	0.385	0.470
2) Real canvas, one family	0.417	0.483	0.348	0.384	0.467
3) Real canvas, all families	0.490	0.579	0.397	0.474	0.534

Swapping the canvas closes the gap. A matched pair isolates the fix: identical checkpoint, shape family (scatter_field), dose, and training budget, only the canvas source changes from the synthetic bank to a real MRI subject. Every source improves except ATLAS v2.0, a single compact chronic lesion – the only non-multi-focal target of the four, plausibly outside what a multiplicity-targeted family like scatter_field can reach.

Extending the real-host canvas from one shape family to all seven, a wider real-subject pool (every organ, CT and MRI), and a longer training budget keeps compounding on GNC_705 – the best result of the whole progression – while easing back slightly on the two brain-lesion sources from stage 2’s peak.

No in-distribution cost. The same checkpoints’ own periodic TotalSegmentator validation shows the OOD gain is additive, not traded against in-distribution accuracy:

The seen/unseen and MRI/CT gaps hold at roughly the same size at every stage; stage 3’s longer budget lifts all five numbers together rather than trading one for another.

Stage 2→3 changes several axes at once (family count, host pool, training budget)

5 Discussion

6 Conclusion

ToDoS

Select the correct language in the first rows of `main.tex`. The % makes a line a comment.

Generate a PDF (Visual Studio Code: click on the green arrow in the top right corner).

Check the PDF (Visual Studio Code: click on the two rectangles with the magnifying glass in the top right corner).

Talk about this template with your supervisor: Should you add a declaration for the use of AI? Currently no official AI-declaration is given. You can find one in the pull requests of the github repository.

With CTRL pressed you can click on the PDF and jump to the corresponding place in your `.tex` files.

Check the Errors. (Visual Studio Code: bottom left) There should be one warning because the first ToDo pushes down the second one: LaTeX Marginpar on page 2 moved.

In the `main.tex` you can press ALT + SHIFT and the ARROW UP and DOWN keys to create a multiline cursor, comment out all the sections you are currently not working on and remove the multiline cursor with ESC.

Keep the following examples and todos till the end and check them out sometimes.

Adjust your License Notes to reflect the used programs.

For Printing: Remove the `todonotes`-package in `setup.tex` to find all Todos.

Table 9: A nice table with a nice caption.

Value & Unit	Value	Unit
9.82 m s ⁻²	9.82	m s ⁻²

For Printing: Choose the correct formatting (search `print` in `setup.tex`).

You can remove the glossary and the symbolslist to make it easier, especially in a Bachelor's Thesis.

6.1 Tables and References

Table 9

This is a cited book [3].

Store and display plots and other graphs as `.pdf` files.

Include Algorithm 1 in the same way you include equations like Equation 4:

$$\begin{aligned}y &= 3x^2 + 5x^2 + 4x + 4x + 7 + 1 \\&= 8x^2 + 8x + 8 \\y &= 8 \cdot (x^2 + x + 1)\end{aligned}\tag{4}$$

6.2 Auto-complete in VSCode

There are also some auto-complete features: type `@` + letter to get the letters greek version.

Type `@v` + letter for the alternative form (like φ for ϕ).

You can also use shortcuts for environments like

$$BAL\tag{5}$$

Type BAL and select the correct option from the autocomplete. Maybe you have to try various ones. The B is the base command and AL is the abbreviation for align.

If you want an environment without an index, you can use BS as base command.

Same for changes to your font, like bold or italic: FIT gives *text*

6.3 ", Quotations and Code

Use " or file-names, file_names and code_snippets.

Algorithm 1: The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in setup.tex.

```
from dataclasses import dataclass
```

```
# https://cd.uni-freiburg.de/farben/
```

```
@dataclass
```

```
class MainColors:
```

```
    blue = "#344A9A"
```

```
    black = "#000000"
```

```
    white = "#FFFFFF"
```

```
@dataclass
```

```
class BackgroundColors:
```

```
    blue_100_percent = "#344A9A"
```

```
    blue_80_percent = "#5D6BAD"
```

```
    blue_60_percent = "#868DC2"
```

```
    blue_80_percent = "#AFB1D8"
```

```
    blue_20_percent = "#D7D8EC"
```

```
    dark_blue = "#00004a"
```

```
    sand_100_percent = "#f6f1e3"
```

```
    sand_50_percent = "#faf8f1"
```

```
@dataclass
```

```
class AdditionalColors:
```

```
    green_100_percent = "#00a082"
```

```
    green_80_percent = "#27b29b"
```

```
    green_60_percent = "#7bc6b4"
```

```
green_80_percent = "#afdace"  
green_20_percent = "#daede7"  
brown_100_percent = "#8f6b30"  
brown_80_percent = "#a58959"  
brown_60_percent = "#bca683"  
brown_80_percent = "#d2c4ac"  
brown_20_percent = "#e9e1d6"  
yellow_100_percent = "#ffe863"  
yellow_80_percent = "#ffed82"  
yellow_60_percent = "#fff1a1"  
yellow_80_percent = "#fff6c1"  
yellow_20_percent = "#fffae0"  
pink_100_percent = "#f5c2ed"  
pink_80_percent = "#f7cef1"  
pink_60_percent = "#f9daf4"  
pink_80_percent = "#fbe7f8"  
pink_20_percent = "#fdf3fb"
```

6.4 Clipboard Manager, Git and more

Clipboard Manager help with copying multiple different things at once. On Windows you can press WINDOWS + V to activate the history and choose older things you copied.

Use Git ([click here](#)).

6.5 Glossary

Click them, the page numbers in the glossaries are also clickable.

name one

name one

name one

long form of name two (short2)

short2

short2

a symbol

a symbol

a symbol

Big Letter at start: Name one

plural of name one

name one

Big Letter at start: Short2

short2

short2s

Big Letter at start: *A symbol*

a symbol

Symbols

Notation	Description	Page List
<i>a symbol</i>	description symbol	22, 23

List of Figures

1	Architecture overview: separate image/label streams + dual attention. .	7
2	Dual attention within one transformer layer, shown for a single context volume ($K=1$ for clarity) and 3 of its N spatial tokens per stream. Left: the modality axis – image and label tokens attend to each other within the same cell. Right: the position axis – tokens attend across cells and, for the target, across contexts. Green edges touch the target row, which cross-attends read-only into context and whose label-stream input is the query prior (subsection 3.3) rather than ground truth.	10
3	Query image prior across two cascade levels, with the full model input/output flow. At level 1 the target has no real prior yet (dashed support-mean placeholder); at level 2 the target’s second lane is a real crop of level 1’s own prediction $\hat{L}_1^t$ – the same lane the context mask always occupies.	11
4	Medverse’s architecture: dual U-Net cross-attention, shown for comparison against our architecture (Figure 1).	IX

List of Tables

1	Positioning against prior in-context segmentation models on the three axes this thesis organizes around.	5
2	Every evaluation source, grouped by relationship to the training vocabulary. n is subjects; n_{cls} is evaluated classes.	13
3	Training settings shared by Dual U-Net Cross-Attn, the single-axis control, and dual attention (ours). Epoch count is a placeholder pending confirmation of the exact per-arm value.	14
4	Dual U-Net Cross-Attn, the single-axis control, and dual attention (ours), across all evaluation datasets (subsection 4.1). Grouped by vocabulary relationship: in-vocabulary seen/unseen, novel-vocabulary (unseen by construction), non-object, held-out modality.	15
5	Predicted-prior checkpoint vs. Medverse (native autoregressive, depth-matched), macro Dice and mean per-sample latency. Levels/spacing is our own cascade ladder for that dataset; Medverse’s AR depth is matched to the same level count.	16
6	Query-prior source ablation, macro Dice, across all validation datasets. TotalSeg CT also reports NSD (0.576 / 0.631).	17
7	OOD Dice, real-host canvas progression. Stage 1→2 is a single-variable comparison (canvas only); stage 2→3 additionally extends to all shape families, a wider real-subject pool, and a longer budget.	18
8	TotalSegmentator validation Dice for the same checkpoints. seen/unseen: macro Dice by trained-class membership. MRI/CT: Dice by target modality.	18
9	A nice table with a nice caption.	21

List of Algorithms

- 1 The Colors from the Corporate Design of the University of Freiburg. The keywords are bold and the comments blue. This can be changed in `setup.tex`. 22

Glossary

Notation	Description	Page List
name one	description one	21, 23
short2	description two	21–23

References

- [1] Benjamin Billot, Douglas N. Greve, Oula Puonti, Axel Thielscher, Koen Van Leemput, Bruce Fischl, Adrian V. Dalca, Juan Eugenio Iglesias, et al. SynthSeg: Segmentation of brain MRI scans of any contrast and resolution without retraining. *Medical Image Analysis*, 86:102789, 2023.
- [2] Victor Ion Butoi, Jose Javier Gonzalez Ortiz, Tianyu Ma, Mert R. Sabuncu, John Guttag, and Adrian V. Dalca. Universeg: Universal medical image segmentation. *arXiv preprint arXiv:2304.06131*, 2023.
- [3] Leander Bürkin. *LaTeX-Template*. Fachschaft TF Freiburg, 2025.
- [4] Özgün Çiçek, Ahmed Abdulkadir, Soeren S. Lienkamp, Thomas Brox, and Olaf Ronneberger. 3D U-net: Learning dense volumetric segmentation from sparse annotation. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2016.
- [5] Jun Fu, Jing Liu, Haijie Tian, Yong Li, Yongjun Bao, Zhiwei Fang, and Hanqing Lu. Dual attention network for scene segmentation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
- [6] Yunhe Gao et al. Show and segment: Universal medical image segmentation via in-context learning. *arXiv preprint*, 2025.
- [7] Ali Hatamizadeh, Yucheng Tang, Vishwesh Nath, Dong Yang, Andriy Myronenko, Bennett Landman, Holger R. Roth, and Daguang Xu. UNETR: Transformers for 3d medical image segmentation. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, 2022.
- [8] others Hu. Medverse: A universal model for medical image segmentation. *arXiv preprint*, 2025.
- [9] Fabian Isensee, Paul F. Jaeger, Simon A. A. Kohl, Jens Petersen, and Klaus H. Maier-Hein. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. *Nature Methods*, 2021.

- [10] Cheng Ouyang, Chen Chen, Surui Li, Zeju Li, Chen Qin, Wenjia Bai, and Daniel Rueckert. Causality-inspired single-source domain generalization for medical image segmentation. *IEEE Transactions on Medical Imaging*, 2022. arXiv:2111.12525.
- [11] Marco Reiser et al. Deep neural patchworks: Coping with large segmentation tasks. In *Medical Image Computing and Computer Assisted Intervention*, 2022.
- [12] Jianlin Su, Yu Lu, Shengfeng Pan, Bo Wen, and Yunfeng Liu. RoFormer: Enhanced transformer with rotary position embedding. *arXiv preprint arXiv:2104.09864*, 2021.
- [13] Keyu Tian, Yi Jiang, Zehuan Yuan, Bingyue Peng, and Liwei Wang. Visual autoregressive modeling: Scalable image generation via next-scale prediction. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [14] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [15] Jakob Wasserthal et al. TotalSegmentator: Robust segmentation of 104 anatomic structures in CT images. *Radiology: Artificial Intelligence*, 2023.

Appendix

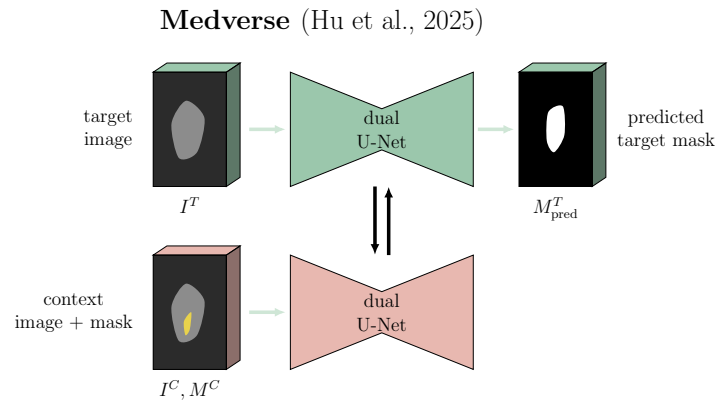


Figure 4: Medverse’s architecture: dual U-Net cross-attention, shown for comparison against our architecture (Figure 1).